

Практикум B5:archive

Every past-paper question on this topic, grouped by technique. Not a practicum — a drill. There is no theory here and no ladder to climb: the theory is in Practicum B5, and this notebook is what you open afterwards, when the only thing left is to do them all until the moves are automatic.

О практикуме

Четвёртый практикум серии на английском, устроенный как B2–B4: ноутбук, карточка приёмов и сообщения проверок английские, документация репозитория остаётся русской.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-b5-exponential-log-models

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

B5 archive: exponential and logarithmic models

Every past-paper question on this topic, grouped by technique. Not a practicum — a drill. There is no theory here and no ladder to climb: the theory is in *Practicum B5*, and this notebook is what you open afterwards, when the only thing left is to do them all until the moves are automatic.

What is inside. The whole of `functions.logarithmic_functions`, `functions.exponential_models`, `functions.mathematical_models` and `number_algebra.exponential_models`, sessions May 2021 — November 2025: **36 questions, 83 marks**, in nine sections, one section per technique.

The corpus records 37 blocks and 85 marks. The one difference is a question filed under `mathematical_models` that has neither an exponential nor a logarithm in it — "*show that $T = 500 \sec \theta + \frac{2500 - 1000 \tan \theta}{3}$* "

", about the time to walk across a beach. Nothing else is left out and nothing is added: this is the topic exactly as the corpus sees it.

Sixteen questions are Paper 1, eight are Paper 2, twelve are Paper 3. The split by paper is also a split by what the answer looks like: Paper 1 wants $\frac{\ln 2}{5730}$ and $3p + q$, Paper 2 and 3 want 2380 and 107. If a Paper 1 answer of yours has a decimal point in it, look again.

How to work. Read the question, answer in the cell below it, run the cell. The check is not a comparison with a stored answer. A model is right when it reproduces the data it was fitted to — `verify_model` puts each pair from the question back into whatever you wrote. A root is right when it satisfies the equation. An expression *in terms of p and q* is right when it uses only those letters and comes out to the right number. So an answer arrived at by a wrong route still has to be true.

Four checks do compare a hash, and they are the same case each time: the answer is a rounding — 2380 years, 10 100 people, 1041 dinar, €5906.23 — and a rounded number satisfies nothing exactly, so there is nothing to substitute it into.

Three questions have no cell at all. 5.1 is a one-mark "*show that* $A_0 = 100$ " where substituting $t = 0$ leaves nothing to check; 8.1 and 8.5 ask you to state an assumption and to comment on two predictions, in words.

Leave a cell blank and it prints  and moves on, which means you can run the whole notebook top to bottom on the first open and nothing breaks.

The solutions are at the end, numbered the same way. Open one after you have worked the question, not before — and read the markscheme note in it, because that is where the marks actually are.

The nine sections.

§	technique	questions	marks
1	The three log laws	3	7
2	Change of base	3	10
3	An equation with the unknown under a logarithm	1	4
4	The laws of exponents, and the relation you were handed	3	5
5	Taking logarithms of both sides	3	7
6	Reading a model, and fitting one	7	15
7	Percentages as a power	5	13
8	The logistic model	5	8
9	The logarithm as a function	6	14

Sections 1–3 move the unknown **up**, out from under a logarithm. Sections 4–8 move it **down**, out of an exponent. Section 9 stops using the logarithm as a tool and starts asking about it as a function.

```

import sys
sys.path.append('.')          # from
practicum/functions to practicum/kit.py
import sympy as sp           # the escape
hatch: anything not in kit is in sp
from kit import *             # checks +
Interval, Union, solveset, Rational

language('en')                # this
notebook is in English, and so are the checks

a, b, n, p, q, r = symbols('a b n p q r')
S, T, v0 = symbols('S T v0', positive=True)
# S here is the sum from question 6.3, and it
# shadows sympy's S. Where a check
# wants "all the reals", write Interval(-oo,
oo) rather than S.Reals.

print('ready; sympy', sp.__version__)
print('in terms of p and q: ', 3*p + q)
print('an exact log:          ', log(2)/5730)
print('a model in t:          ',
15000*exp(log(Rational(89, 100))/8*t))
print('a set of values:      ',
Interval(exp(-5), E))
print('a list of roots:      ', [Rational(1,
25)])

```

1. The three log laws

$\log_b MN = \log_b M + \log_b N$, $\log_b \frac{M}{N} = \log_b M - \log_b N$, $\log_b M^r = r \log_b M$ — and there is no law for $\log_b(M + N)$. The loose number

next to a logarithm has to become one:
 $1 = \log_b b$.

1.1 — May 2025 TZ3 Paper 1 Q1(a), 3 marks

Let $\log_{10} 2 = p$ and $\log_{10} 3 = q$. Find an expression for $\log_{10} 24$ in terms of p and q .

```
q1_1 = ...          # log_10 24, in terms of p
and q

verify_in_terms_of('1.1', q1_1, log(24, 10),
{p: log(2, 10), q: log(3, 10)})
```

1.2 — May 2024 TZ1 Paper 1 Q2(a), 2 marks

It is given that $\log_{10} a = \frac{1}{3}$, where $a > 0$.
Find the value of $\log_{10} \frac{1}{a}$.

```
q1_2 = ...          # log_10 (1/a)

verify_exact('1.2', q1_2, -Rational(1, 3))
```

1.3 — May 2025 TZ1 Paper 1 Q8(a), 2 marks

Show that $1 + \log_2 n \geq \log_2(n + 1)$ for $n \in \mathbb{Z}^+$.

The answer is printed in the question, so the cell checks the step that earns the marks: write $1 + \log_2 n$ as a **single logarithm**.

```
q1_3 = ...      # 1 + log_2 n as one
logarithm, e.g. log(<something>, 2)

verify_identity('1.3', q1_3, 1 + log(n, 2),
var=n)
```

2. Change of base

$\log_b a = \frac{\log_c a}{\log_c b}$ — the argument on top, the base underneath. Until two logarithms share a base, no law from section 1 applies to them. When the bases are powers of one number, $\log_{b^m} a = \frac{1}{m} \log_b a$ does it in one line.

2.1 — *May 2025 TZ3 Paper 1 Q1(b), 2 marks*

With $\log_{10} 2 = p$ and $\log_{10} 3 = q$, find an expression for $\log_3 8$ in terms of p and q .

```
q2_1 = ...          # log_3 8, in terms of p and q
```

```
verify_in_terms_of('2.1', q2_1, log(8, 3),  
{p: log(2, 10), q: log(3, 10)})
```

2.2 — May 2024 TZ1 Paper 1 Q2(b), 3 marks

It is given that $\log_{10} a = \frac{1}{3}$, where $a > 0$.
Find the value of $\log_{1000} a$.

```
q2_2 = ...          # log_1000 a
```

```
verify_exact('2.2', q2_2, Rational(1, 9))
```

2.3 — November 2025 TZ3 Paper 1 Q4, 5 marks

Solve the equation $3\log_8 10x - \log_4 x = 1$
for $x > 0$.

Enter a **list** of roots: `[Rational(1, 5)]`, `[4, 9]`.

```
q2_3 = [...]        # all roots with x > 0
```

```
verify_roots('2.3', q2_3, 3*log(10*x, 8) -  
log(x, 4) - 1, (0.0001, 100))
```

3. An equation with the unknown under a logarithm

Bases first, then collapse each side to one logarithm, then remove them — and then check that every argument was positive at the root you found. Removing logarithms is not reversible, and it invents roots.

3.1 — November 2025 TZ1 Paper 1 Q2(b), 4 marks

Part (a) of this question established that for $x > 7$,

$$\frac{x}{x^2 - 8x + 7} \times \frac{x^2 - 1}{x + 1} \equiv \frac{x}{x - 7}.$$

Hence, or otherwise, solve

$$\begin{aligned} \log_2[x(x^2 - 1)] - 1 \\ = \log_2[(x^2 - 8x + 7)(x + 1)]. \end{aligned}$$

```
q3_1 = [...]      # all roots

verify_root_set('3.1', q3_1,
                 Eq(log(x*(x**2 - 1), 2) - 1,
                    log((x**2 - 8*x + 7)*(x + 1), 2)),
                 domain=Interval.open(7, oo))
```

4. The laws of exponents, and the relation you were handed

$a^m a^n = a^{m+n}$ and $(a^m)^n = a^{mn}$, used not to tidy an answer but to make a substitution possible. And $c = e^{\ln c}$ for every $c > 0$, which is the definition of $\ln$ read backwards.

4.1 — May 2021 TZ2 Paper 1 Q11(c), 2 marks

A particle has velocity $v(t) = (1 + v_0)e^{-t} - 1$, and the time T at which its displacement is greatest satisfies $e^T = 1 + v_0$. By using that result, show that $v(T - k) = e^k - 1$.

The answer is printed in the question, so the cell checks the substitution the marks are for: write e^{-T} in terms of v_0 .

```
q4_1 = ...          # e^(-T), in terms of v0

verify_identity('4.1', q4_1, 1/(1 + v0),
var=v0)
```

4.2 — May 2021 TZ2 Paper 1 Q11(d), 2 marks

Similarly, let $v(T + k)$ represent the particle's velocity k seconds after it reaches its maximum displacement.

Deduce a similar expression for $v(T + k)$ in terms of k .

```
q4_2 = ...          # v(T + k), in terms of k
```

```
verify_exact('4.2', q4_2, exp(-k) - 1)
```

4.3 — May 2025 TZ1 Paper 1 Q7(a), 1 mark

Consider the complex number $z = 3^{i-1}$. Write the integer 3 in the form e^a , where $a \in \mathbb{R}$.

```
q4_3 = ...          # a
```

```
verify_exact('4.3', q4_3, log(3))
```

5. Taking logarithms of both sides

Isolate the power first, then take logarithms, then bring the exponent

down. Keep the exact form to the last line: rounding k and then using it moves the answer by more than the rounding.

5.1 — May 2021 TZ2 Paper 2 Q5(a), 1 mark

The amount A of carbon-14 present in a plant t years after its death is modelled by $A = A_0 e^{-kt}$, $t \geq 0$, with $A_0, k > 0$. At the time of death a plant is defined to have 100 units of carbon-14.

Show that $A_0 = 100$.

One mark, and once you have put $t = 0$ there is nothing left to check — so this question has no cell. Write the line down anyway; it is a mark.

5.2 — May 2021 TZ2 Paper 2 Q5(b), 3 marks

The time taken for half the original amount of carbon-14 to decay is known to be 5730 years. Show that $k = \frac{\ln 2}{5730}$.

The answer is printed in the question; enter it as an exact expression so the

check can confirm that it is what your working produced.

```
q5_2 = ...          # k, exactly  
  
verify_exact('5.2', q5_2, log(2)/5730)
```

5.3 — May 2021 TZ2 Paper 2 Q5(c), 3 marks

Find, correct to the nearest 10 years, the time taken after the plant's death for 25% of the carbon-14 to decay.

```
q5_3 = ...          # the time, to the nearest  
10 years  
  
check_num('5.3', q5_3, 3, 'b56267433ab0')
```

6. Reading a model, and fitting one

The first six questions here put numbers into a model somebody else built — which is what fitting starts with. The last one builds it: two constants, two conditions, and one decision about what $t = 0$ means.

6.1 — May 2023 TZ1 Paper 2 Q3(a), 2 marks

The total number of children y visiting a park depends on the highest temperature T in degrees Celsius. A park official predicts it with $y = -0.6T^2 + 23T + 110$, where $10 \leq T \leq 35$.

Use this model to estimate the number of children in the park on a day when the highest temperature is 25°C .

```
q6_1 = ...          # the number of children  
  
verify_exact('6.1', q6_1, 310)
```

6.2 — November 2025 TZ1 Paper 2 Q10(a), 2 marks

An airplane lands on a runway 100 metres in front of a stationary car; at that instant the car starts to travel in the same direction. For $t \geq 0$ seconds after landing, the velocities in m s^{-1} are

$$v_{\text{air}} = 60e^{-0.1t}, \quad v_{\text{car}} = 5t.$$

When the airplane lands, write down the speed of **(i)** the airplane; **(ii)** the car.

```
q6_2a = ...      # the airplane
q6_2b = ...      # the car

verify_exact('6.2a', q6_2a, 60)
verify_exact('6.2b', q6_2b, 0)
```

6.3 — May 2023 TZ1 Paper 3 Q2(c), 1 mark

Let $M_n(S)$ be the maximum product of n positive real numbers whose sum is S .

For $n = 2$ this product is $M_2(S) = \left(\frac{S}{2}\right)^2$.

Verify that $M_2(S) = \left(\frac{S}{2}\right)^2$ is true for $S = 12$ — that is, give the value it produces.

```
q6_3 = ...      # M_2(12)

verify_exact('6.3', q6_3, 36)
```

6.4-6.6 — May 2023 TZ1 Paper 3 Q2(e), 1 mark each

It has been proved that $M_n(S) = \left(\frac{S}{n}\right)^n$.

Hence determine **(i)** $M_3(12)$; **(ii)** $M_4(12)$; **(iii)** $M_5(12)$.

All three are exact.

```

q6_4 = ...      # M_3(12)
q6_5 = ...      # M_4(12)
q6_6 = ...      # M_5(12)

verify_exact('6.4', q6_4, 4**3)
verify_exact('6.5', q6_5, 3**4)
verify_exact('6.6', q6_6, Rational(12, 5)**5)

```

6.7 — November 2022 Paper 2 Q4, 7 marks

The population of a town t years after 1 January 2014 can be modelled by

$$P(t) = 15\,000e^{kt}, \quad k < 0, t \geq 0.$$

It is known that between 1 January 2014 and 1 January 2022 the population decreased by 11%.

(a) Write down the completed model, with k in place.

(b) Use it to estimate the population of this town on 1 January 2041.

The first check puts $t = 0$ and $t = 8$ back into whatever you wrote and asks whether 15 000 and 13 350 come out. It says nothing about 2041.

```
q6_7a = ...      # the model, an expression
in t with no letters left in it
q6_7b = ...      # the population on 1
January 2041

verify_model('6.7a', q6_7a, [(0, 15000), (8,
13350)])
check_num('6.7b', q6_7b, 3, 'a91bb59c3f9c')
```

7. Percentages as a power

A percentage change per period is a multiplier per period, and the exponent counts **periods**, not years. Nominal 4% compounded quarterly is 1% four times. Inflation at j against interest at i leaves $\frac{1+i}{1+j}$ per period — and the markscheme also accepts the approximation $1 + i - j$.

7.1 — *May 2025 TZ1 Paper 1 Q2(a), 1 mark*

Bob invests 1000 dinar in an account paying a nominal annual interest rate of 4% compounded **quarterly**. The amount after one complete year can be written as $1000(1+k)^4$, where $k \in \mathbb{Q}$. Write down the value of k .

```
q7_1 = ...          # k, exactly  
  
verify_exact('7.1', q7_1, Rational(1, 100))
```

7.2 — May 2025 TZ1 Paper 1 Q2(c), 4 marks

Hence or otherwise, find the amount of money in the account after one complete year, giving your answer correct to the nearest dinar.

No calculator: part (b) of that question expanded $(1 + x)^4$ for you.

```
q7_2 = ...          # the amount, to the nearest  
dinar  
  
check_num('7.2', q7_2, 4, '8c6c42f379f0')
```

7.3 — May 2025 TZ2 Paper 2 Q4(a), 2 marks

Alex purchases a car for €30 000. The value of the car depreciates at 15% per annum. Find the value of the car after ten years, giving your answer to two decimal places.

```
q7_3 = ...          # the value after ten years  
  
check_num('7.3', q7_3, 6, '774db6f92305')
```

7.4 — May 2025 TZ2 Paper 2 Q4(b), 4 marks

Alex invests €50 000 in a bank account that pays a compound interest rate of 1.5% per month. Inflation over the same period was 0.8% per month. Find the number of months required for the **real** value of the investment to first exceed € 55 000.

```
q7_4 = ...          # the number of months  
  
verify_exact('7.4', q7_4, 14)
```

7.5 — November 2025 TZ1 Paper 1 Q5(b), 2 marks

The first four terms of the Maclaurin expansion of $(1 - x)^{-4}$ are $1 + 4x + 10x^2 + 20x^3$.

A car was purchased four years ago, depreciated in value by 10% each year, and is worth \$1000 today. Using that

expansion, estimate the value of the car four years ago.

```
q7_5 = ...          # the value four years ago  
  
verify_exact('7.5', q7_5, 1520)
```

8. The logistic model

$x = \frac{L}{1 + Ce^{-kt}}$. L is read off the page, C comes from $t = 0$ where $e^0 = 1$, and k comes from the second data point by section 5.

8.1 — November 2025 TZ1 Paper 3 Q2(a)(i), 1 mark

A wolf population had a stable size of 200. After some years of disruption it fell to 40 ; the area then became protected and the population began to grow again. Researchers model its size x after t years of protection with

$$x = \frac{L}{1 + Ce^{-kt}}, \quad L, C, k \in \mathbb{R}^+,$$

and decide to let $L = 200$.

State the assumption being made in assuming $L = 200$.

*A sentence, so this question has no cell.
The answer is in the solutions.*

8.2 — November 2025 TZ1 Paper 3 Q2(a)(ii), 2 marks

At $t = 0$ the population of wolves is 40.
Find the value of C .

```
q8_2 = ...          # C  
  
verify_exact('8.2', q8_2, 4)
```

8.3 — November 2025 TZ1 Paper 3 Q2(a)(iii), 2 marks

At $t = 5$ the population has increased to 70. Find the value of k .

The paper asks for three significant figures; enter the exact form and the check will accept it.

```
q8_3 = ...          # k, exactly  
  
verify_exact('8.3', q8_3, log(Rational(28,  
13))/5)
```

8.4 — November 2025 TZ1 Paper 3

Q2(a)(iv), 2 marks

(a) Write down the completed model.

(b) Use it to predict the size of the wolf population 10 years after the area became protected, correct to the nearest whole number.

```
q8_4a = ...      # x(t), an expression in t
                  with no letters left in it
q8_4b = ...      # the population at t = 10

verify_model('8.4a', q8_4a, [(0, 40), (5,
70)])
verify_exact('8.4b', q8_4b, 107)
```

8.5 — November 2025 TZ1 Paper 3

Q2(c), 1 mark

An alternative model — the Gompertz model, which satisfies $\frac{dx}{dt} = ax \ln \frac{200}{x}$ — predicts 101 wolves at $t = 10$. After 10 years the population is measured and found to be 85.

Comment on the predictions made by the two models.

A sentence, so this question has no cell either.

9. The logarithm as a function

Not a tool now but a function: inside a composition, defining a domain, or serving as a measuring scale. Solve for $\ln x$ first and remove the logarithm last — e^u is increasing, so it never flips an inequality.

9.1 — May 2024 TZ2 Paper 2 Q3(a), 1 mark

The loudness L of a sound, in decibels, is related to its intensity I units by $L = 10 \log_{10}(I \times 10^{12})$. Sound S_1 has intensity 10^{-6} units and loudness 60 decibels; sound S_2 has twice the intensity of S_1 .

State the intensity of S_2 .

```
q9_1 = ...          # the intensity of S2

verify_exact('9.1', q9_1, 2*Rational(1,
10**6))
```

9.2 — May 2024 TZ2 Paper 2 Q3(b), 2 marks

Determine the loudness of S_2 .

The markscheme accepts the exact form as a final answer, and so does the check.

```
q9_2 = ...          # the loudness of S2  
  
verify_exact('9.2', q9_2, 60 + 10*log(2, 10))
```

9.3 — May 2024 TZ2 Paper 2 Q3(c), 3 marks

The maximum loudness of thunder in a thunderstorm was measured to be 115 decibels. Find the corresponding intensity I of the thunder.

```
q9_3 = ...          # the intensity of the  
thunder  
  
verify_exact('9.3', q9_3, 10**Rational(-1,  
2))
```

9.4 — May 2025 TZ1 Paper 1 Q10(d), 3 marks

The function f is defined by $f(x) = 5(x + 1)(x + 3)$, $x \in \mathbb{R}$; the function g by $g(x) = \ln x$, $x > 0$. An earlier part established that $f(x) \leq 40 \iff -5 \leq x \leq 1$.

(a) Write down an expression for $(f \circ g)(x)$.

(b) Solve $(f \circ g)(x) \leq 40$.

```
q9_4a = ...      # (f o g)(x)
q9_4b = ...      # the solution set

verify_identity('9.4a', q9_4a, 5*(log(x) +
1)*(log(x) + 3))
verify_solution_set('9.4b', q9_4b, 5*(log(x)
+ 1)*(log(x) + 3) <= 40,
                    domain=Interval.open(0,
oo))
```

9.5 — May 2025 TZ1 Paper 1 Q10(e), 3 marks

With the same f and g , find the domain of $g \circ f$.

```
q9_5 = ...      # the domain of g o f

verify_solution_set('9.5', q9_5, 5*(x + 1)*(x
+ 3) > 0)
```

9.6 — May 2023 TZ1 Paper 3 Q2(i), 2 marks

The function g is defined by $\ln(g(x)) = x \ln \frac{S}{x}$, for $x \in \mathbb{R}^+$. Find $g(x)$, and so verify that $g(x) = M_x(S)$ when $x \in \mathbb{Z}^+$.

```
q9_6 = ...          # g(x), in terms of S and x  
  
verify_identity('9.6', q9_6, exp(x*log(S/x)),  
var=x)
```

Solutions

Numbered as the questions are. Each names its source and the marks it carries.

1. The three log laws

1.1 — *May 2025 TZ3 P1 Q1(a), 3 marks.*

Factorise first, because there is no law for a sum:

$$\begin{aligned}24 &= 2^3 \times 3 \\&\implies \log_{10} 24 \\&= 3 \log_{10} 2 + \log_{10} 3 \\&= 3p + q.\end{aligned}$$

(M1) for using a law, **(A1)** for reaching $3 \log_{10} 2 + \log_{10} 3$ with the arguments actually equal to 2 and 3, **A1** for $3p + q$.

1.2 — *May 2024 TZ1 P1 Q2(a), 2 marks.*

$$\log_{10} \frac{1}{a} = \log_{10} 1 - \log_{10} a = -\log_{10} a = -\frac{1}{3}.$$

Equally $\log_{10} a^{-1} = -\log_{10} a$. Note $\log_{10} 1 = 0$, not 1.

1.3 — *May 2025 TZ1 P1 Q8(a), 2 marks.*

The loose 1 joins the logarithm:

$$1 + \log_2 n = \log_2 2 + \log_2 n = \log_2(2n).$$

Then $2n \geq n + 1$ for $n \geq 1$, and since $\log_2$ **is increasing**, $\log_2(2n) \geq \log_2(n + 1)$. The second half is the **R1**, and it is for naming monotonicity out loud — the inequality $2n \geq n + 1$ alone does not earn it. Part (b) of that question then runs induction on this result to prove $n > \log_2 n$.

2. Change of base

2.1 — *May 2025 TZ3 P1 Q1(b), 2 marks.*

$$\log_3 8 = \frac{\log_{10} 8}{\log_{10} 3} = \frac{3 \log_{10} 2}{\log_{10} 3} = \frac{3p}{q}.$$

Argument on top, base underneath. Check the direction once on something known — $\log_2 8 = \frac{\log 8}{\log 2} = 3$ — and you will not flip it again.

2.2 — *May 2024 TZ1 P1 Q2(b), 3 marks.*

$$\log_{1000} a = \frac{\log_{10} a}{\log_{10} 1000} = \frac{1/3}{3} = \frac{1}{9}.$$

The 3 underneath is $\log_{10} 10^3$. The markscheme prints four routes to the

same place, including $10^{1/3} = 1000^x = (10^3)^x$ and hence $3x = \frac{1}{3}$.

2.3 — *November 2025 TZ3 P1 Q4, 5 marks.* Everything is a power of 2:

$$\begin{aligned} 3 \log_8 10x &= 3 \cdot \frac{\log_2 10x}{3} \\ &= \log_2 10x, & \log_4 x \\ &= \frac{\log_2 x}{2} \\ &= \log_2 \sqrt{x}. \end{aligned}$$

So

$$\begin{aligned} \log_2 \frac{10x}{\sqrt{x}} &= \log_2 2 \\ &\implies 10\sqrt{x} \\ &= 2 \\ &\implies \sqrt{x} \\ &= \frac{1}{5} \\ &\implies x \\ &= \frac{1}{25}. \end{aligned}$$

The two **M** marks — power/quotient rule and change of base — are independent and may be awarded in any order, so a wrong order still scores. The markscheme prints $10\sqrt{x} = 2$ as its own

A1, which is a warning: stopping at $\sqrt{x} = \frac{1}{5}$ answers a different question.

3. An equation with the unknown under a logarithm

3.1 — *November 2025 TZ1 P1 Q2(b), 4 marks.* The -1 is the question. Either move it across as $1 = \log_2 2$, or absorb it as $\log_2 \frac{1}{2}$. Collapsing both sides,

$$\log_2 \frac{x(x^2 - 1)}{(x^2 - 8x + 7)(x + 1)} = \log_2 2.$$

The fraction inside is exactly what part (a) simplified, so it equals $\frac{x}{x-7}$, and removing the logarithms gives

$$\frac{x}{x-7} = 2 \implies x = 2x - 14 \implies x = 14.$$

$14 > 7$, so every argument is positive and the root stands.

The markscheme's METHOD 2 does it the other way round — use $1 = \log_2 2$ on the left first, get $\log_2 \frac{x^3 - x}{2} = \log_2 ((x^2 - 8x + 7)(x + 1))$, and cancel — and lands on $\frac{x}{2} = x - 7$. Same answer, same marks.

Why the domain is not decoration.

Removing logarithms is not reversible: any root it produced below 7 would have been manufactured by that step and not present in the original equation. Here none was, but the habit is what the check is testing.

4. The laws of exponents

4.1 — *May 2021 TZ2 P1 Q11(c), 2 marks.* $e^T = 1 + v_0$ gives $e^{-T} = \frac{1}{1 + v_0}$ — the reciprocal, and this is the line the two marks rest on. Then split the exponent:

$$\begin{aligned}v(T - k) &= (1 + v_0)e^{-(T-k)} - 1 \\&= (1 + v_0)e^{-T}e^k - 1 \\&= (1 + v_0) \cdot \frac{1}{1 + v_0} \cdot e^k - 1 \\&= e^k - 1.\end{aligned}$$

The markscheme's METHOD 2 is shorter: write $1 + v_0$ as e^T from the start, and $e^T e^{-(T-k)} = e^k$ falls out in one step.

4.2 — *May 2021 TZ2 P1 Q11(d), 2 marks.* *Deduce* means do not start again. Replacing k by $-k$ turns " k seconds before" into " k seconds after": $v(T+k) = e^{-k} - 1$. Part (e) then adds the two: $e^k + e^{-k} - 2 \geq 0$ by AM-GM, so the particle is faster before the turning point than it is slow after it.

4.3 — *May 2025 TZ1 P1 Q7(a), 1 mark.* $3 = e^{\ln 3}$, so $a = \ln 3$. Not a trick — the definition of $\ln$ read backwards. The rest of that question uses it to make sense of $3^{i-1} = e^{(i-1)\ln 3}$ and to extract $\operatorname{Re}(z) = \frac{1}{3} \cos(\ln 3)$.

5. Taking logarithms of both sides

5.1 — *May 2021 TZ2 P2 Q5(a), 1 mark.* $A(0) = A_0 e^0 = A_0$, and the plant has 100 units at death, so $A_0 = 100$. The markscheme wants the line $100 = A_0 e^0$ written down; the conclusion alone is **A0**.

5.2 — *May 2021 TZ2 P2 Q5(b), 3 marks.* Half the original amount after 5730 years:

$$\begin{aligned}
50 &= 100e^{-5730k} \\
&\implies e^{-5730k} \\
&= \frac{1}{2} \\
&\implies e^{5730k} \\
&= 2 \\
&\implies 5730k \\
&= \ln 2 \\
&\implies k \\
&= \frac{\ln 2}{5730}.
\end{aligned}$$

The route through $e^{5730k} = 2$ avoids $\ln \frac{1}{2} = -\ln 2$, which is the most common sign slip in this topic. The markscheme is unusually generous here — *"award full marks for at least two correct algebraic steps seen"* — because several routes exist.

5.3 — *May 2021 TZ2 P2 Q5(c), 3 marks.*
 25% decayed means 75 units **remain**:

$$\begin{aligned}
75 &= 100e^{-\frac{\ln 2}{5730}t} \\
&\implies \ln 0.75 \\
&= -\frac{\ln 2}{5730}t \\
&\implies t \\
&= \frac{5730 \ln \frac{4}{3}}{\ln 2} \\
&= 2378.16 \dots
\end{aligned}$$

To the nearest 10 years, $t = 2380$. This is the one question in the topic where the markscheme prints both routes and pays the same for either: **EITHER** solve graphically **OR** manipulate logs. What the graph does not give you is the exact form — which costs nothing here and costs the whole mark in part (b).

6. Reading a model, and fitting one

6.1 — *May 2023 TZ1 P2 Q3(a), 2 marks.*

$y = -0.6(25)^2 + 23(25) + 110 = -375 + 575 + 110 = 310$ children. Exact, despite being a calculator paper: the model is a quadratic with terminating coefficients.

6.2 — *November 2025 TZ1 P2 Q10(a), 2 marks.* At $t = 0$: $v_{\text{air}} = 60e^0 = 60 \text{ m s}^{-1}$ and $v_{\text{car}} = 5 \times 0 = 0$. One mark each, and the whole content is $e^0 = 1$. The rest of that question integrates the difference to find when the car catches the plane.

6.3 — *May 2023 TZ1 P3 Q2(c), 1 mark.* $M_2(12) = \left(\frac{12}{2}\right)^2 = 6^2 = 36$, which agrees with parts (a) and (b): the two numbers summing to 12 with the largest product are 6 and 6.

6.4-6.6 — *May 2023 TZ1 P3 Q2(e), 1 mark each.*

$$M_3(12) = 4^3 = 64,$$

$$M_4(12) = 3^4 = 81,$$

$$M_5(12) = \left(\frac{12}{5}\right)^5 = 2.4^5 = 79.62624.$$

The sequence is the point: the product rises, peaks between $n = 4$ and $n = 5$, and falls. Later parts of that question find the peak exactly — it is at $x = S/e$, which for $S = 12$ is 4.41, and it is why $n = 4$ beats $n = 5$.

6.7 — *November 2022 Paper 2 Q4, 7 marks.* $t = 0$ is 1 January 2014, so 1 January 2022 is $t = 8$ and 1 January 2041 is $t = 27$. A decrease of 11% leaves 89%:

$$P(8) = 0.89 \times 15$$

$$000 = 13$$

$$350 \implies e^{8k} = 0.89 \implies k = \frac{\ln 0.89}{8} = -0.014566\dots$$

Then $P(27) = 15\,000e^{-0.014566\dots \times 27} = 10\,122.3\dots$, and the markscheme accepts 10 100 (three significant figures) or 10 122.

Where the seven marks are:
 recognising $P(0) = 15\,000$; computing $0.89 \times 15\,000$;
 recognising $t = 8$;
 substituting both; finding k ; substituting $t = 27$; the number. Six of the seven are before the exponential is ever evaluated, and the two most often lost are the third and the sixth — the two that are only about what t means. Answering with $t = 19$ gives 11 400 and scores five of the seven.

7. Percentages as a power

7.1 — *May 2025 TZ1 P1 Q2(a), 1 mark.*

Nominal 4% compounded quarterly is $\frac{4\%}{4} = 1\%$ per quarter, so $k = \frac{4}{400} = \frac{1}{100}$. Everything downstream depends on this one mark.

7.2 — *May 2025 TZ1 P1 Q2(c), 4 marks.*

No calculator, so expand rather than evaluate. With $(1+x)^4 = 1 + 4x + 6x^2 + 4x^3 + x^4$ at $x = \frac{1}{100}$,

$$\begin{aligned} 1000 \left(1 + \frac{1}{100}\right)^4 &= 1000 \\ &+ 40 \\ &+ 0.6 \\ &+ 0.004 \\ &+ \dots \\ &= 1040.6 \dots \end{aligned}$$

so 1041 dinar. The first two terms already give 1040; the third decides the rounding. The markscheme's METHOD 2 multiplies 1.01 out by hand — $(1.0201)(1.030301)$ — and is longer.

7.3 — *May 2025 TZ2 P2 Q4(a), 2 marks.*

Depreciation at 15% is a multiplier of 0.85:

$$30\,000 \times 0.85^{10} = 5906.2321 \dots \rightarrow \text{€}5906.23.$$

Two decimal places, because it is money and because the question says so.

7.4 — *May 2025 TZ2 P2 Q4(b), 4 marks.*

The exact real multiplier per month is the quotient

$$\begin{aligned} \frac{1.015}{1.008} &= 1.00694 \dots \\ \implies n &> \frac{\ln 1.1}{\ln 1.00694 \dots} \\ &= 13.7722 \dots \\ \implies n & \\ &= 14. \end{aligned}$$

The markscheme's own METHOD 1 is the approximation $r = 1.5 - 0.8 = 0.7\%$, giving $n > \frac{\ln 1.1}{\ln 1.007} = 13.6633 \dots$ and the same answer, for full marks; with it, $n = 13$ gives €54 746.09 and $n = 14$ gives €55 129.31

.

The wrong answer the markscheme names. 164 months, worth **(A1)(M1)** **(A0)(A0)** — the method mark survives, both answer marks are gone. It comes from reading 0.7% as an *annual* rate and

compounding it monthly: $\frac{\ln 1.1}{\ln(1+0.007/12)} = 163.5\dots$ The period was in the question all along.

7.5 — *November 2025 TZ1 P1 Q5(b), 2 marks.* Four years of 10% depreciation multiply by 0.9^4 , so the value four years ago is $1000 \times 0.9^{-4} = 1000(1 - 0.1)^{-4}$. With the given expansion at $x = 0.1$,

$$\begin{aligned}(1 - x)^{-4} &\approx 1 \\ &\quad + 4(0.1) \\ &\quad + 10(0.01) \\ &\quad + 20(0.001) \\ &= 1.52 \\ &\implies \$1520.\end{aligned}$$

The exact value is $1000/0.6561 = 1524.16\dots$; the series truncated at x^3 gives 1520, which is what the question asks for. Part (a) of that question is where $a = 4$ and $b = 10$ come from, by differentiating the Maclaurin series.

8. The logistic model

8.1 — *November 2025 TZ1 P3 Q2(a)(i), 1 mark.* **The assumption is that the long-term population after the disruption will be the same as it was before** — that the carrying capacity has not changed. The markscheme condones "*the maximum value / carrying capacity of the population will be 200*".

8.2 — *...(ii), 2 marks.* At $t = 0$, $e^0 = 1$:

$$\begin{aligned}40 &= \frac{200}{1 + C} \\ \implies 40 + 40C &= 200 \\ \implies C &= 4.\end{aligned}$$

Forgetting $e^0 = 1$ is the only way to lose this.

8.3 — *...(iii), 2 marks.* At $t = 5$:

$$\begin{aligned}
70 &= \frac{200}{1 + 4e^{-5k}} \\
&\implies 1 + 4e^{-5k} \\
&= \frac{20}{7} \\
&\implies e^{-5k} \\
&= \frac{13}{28} \\
&\implies k \\
&= \frac{1}{5} \ln \frac{28}{13} \\
&= 0.153451 \dots \rightarrow 0.153.
\end{aligned}$$

The markscheme prints it as $-\frac{1}{5} \ln \frac{130}{280}$ — the same number with the minus still inside.

8.4 — ...(*iv*), 2 marks. $x(t) = \frac{200}{1 + 4e^{-0.153451 \dots t}}$ and

$$\begin{aligned}
x(10) &= \frac{200}{1 + 4e^{-1.53451 \dots}} \\
&= 107.397 \dots \rightarrow 107 \text{ wolves.}
\end{aligned}$$

Whole animals, so a whole number.

The ten-second sanity check. Evaluate at a large t : the model must approach 200 from below. With the sign of k flipped it collapses towards zero instead —

invisible in the algebra, obvious in the number.

8.5 — ...(*c*), 1 mark. Both models overestimate, and the Gompertz model is closer to the true value.

Logistic 107, Gompertz 101, measured 85. The markscheme accepts either half of that sentence and follows through on your own two numbers, as long as both are positive and both are compared with 85.

9. The logarithm as a function

9.1 — May 2024 TZ2 P2 Q3(a), 1 mark.

Twice 10^{-6} is 2×10^{-6} , that is $\frac{1}{500\,000}$. A mark for reading.

9.2 — ...(*b*), 2 marks.

$$\begin{aligned} L &= 10 \log_{10}(2 \times 10^{-6} \times 10^{12}) \\ &= 10 \log_{10}(2 \times 10^6) \\ &= 10(6 + \log_{10} 2) \\ &= 60 + 10 \log_{10} 2 \\ &= 63.0102 \dots \rightarrow 63.0. \end{aligned}$$

The markscheme accepts $60 + 10 \log_{10} 2$ as a final answer. Notice what did not happen: doubling the intensity did not double the loudness, it added about 3 decibels. That is what a logarithmic scale is *for* — it turns multiplication into addition. The markscheme also names the wrong answer $L = 0$, which comes from using $I = 10^{-12}$.

9.3 — ...*(c)*, 3 marks.

$$\begin{aligned}
 115 &= 10 \log_{10}(I \times 10^{12}) \\
 &\implies I \times 10^{12} \\
 &= 10^{11.5} \\
 &\implies I \\
 &= 10^{-0.5} \\
 &= \frac{1}{\sqrt{10}} \\
 &= 0.316227 \dots
 \end{aligned}$$

Exact forms $10^{-0.5}$ and $\frac{1}{\sqrt{10}}$ are both accepted.

9.4 — May 2025 TZ1 P1 Q10(d), 3 marks.

$(f \circ g)(x) = f(\ln x) = 5(\ln x + 1)(\ln x + 3)$; the markscheme also accepts $5(\ln x + 2)^2 - 5$ and $5(\ln x)^2 + 20 \ln x + 15$.

Then, using the earlier result $f(x) \leq 40 \iff -5 \leq x \leq 1$ with x replaced by $\ln x$,

$$-5 \leq \ln x \leq 1 \implies e^{-5} \leq x \leq e,$$

because e^u is increasing and so preserves both inequalities. The **(M1)** is precisely for "*attempt to replace x with $\ln x$ using their solution to part (c)*" — that is, for **not** solving the quadratic again. Applying e to -5 and writing $-e^5$ is the standard error.

9.5 — $\dots(e)$, 3 marks. $(g \circ f)(x) = \ln(f(x))$ needs $f(x) > 0$:

$$\begin{aligned}(x+1)(x+3) &> 0 \\ \implies x &< \\ &-3 \text{ or } x > \\ &-1,\end{aligned}$$

that is $(-\infty, -3) \cup (-1, \infty)$. Two separate marks: one for the critical values, one for the inequalities pointing the right way. Writing $-3 < x < -1$ — the inside of the parabola, where f is negative — collects the first and loses the second. Note that the domain of the composition is

narrower than the domain of f , which is all of $\mathbb{R}$.

9.6 — *May 2023 TZ1 P3 Q2(i), 2 marks.*

Use the power law on the right and then exponentiate:

$$\begin{aligned}\ln(g(x)) &= x \ln \frac{S}{x} \\ &= \ln \left(\frac{S}{x} \right)^x \\ &\implies g(x) \\ &= \left(\frac{S}{x} \right)^x,\end{aligned}$$

which is $M_x(S)$ whenever x is a positive integer. **M1** for moving the x up as a power, **A1** for g itself.

Why anyone would bother. $M_n(S)$ is defined only for whole n — you cannot have 4.41 factors. Writing it as g , a function of a real variable that agrees with M_n at the integers, is what makes it differentiable, and differentiating is how the question then finds the best n . It is the same move that turns compound interest into e^{rt} .

What to do when it goes wrong

Your model gives the wrong number at a data point. The constants are not the problem — the time is. Write down what $t = 0$ means and recount.

Your answer is off by a factor you cannot find. Check whether a percentage went in as 0.11 where 0.89 was meant, or as 1.04 where 1.01 was meant, or whether the exponent counted years where it should have counted months.

Your logarithmic equation has a root the check rejects. Removing logarithms is not reversible. Put the root back into the *original* equation and look at every argument.

Your inequality came out backwards. You applied e^u or $\ln$ to one side only, or to a negative number. Both are increasing: they never flip an inequality, and they never accept a negative argument.

Your exact answer became a decimal.

On Paper 1 that is the whole mark. Keep $\ln 2$, keep $\frac{1}{9}$, keep $3p + q$, keep $10^{-1/2}$. The calculator comes out only when the paper number says it may.

Nothing is wrong and you are still slow. Sections 1-3 should take seconds, not minutes. If they do not, the three laws are not yet automatic, and no amount of model-fitting practice will fix that.